TAP-N-PRINT : QR BASED SMART PRINTING KIOSK

Anitha R
Asst. Prof., Dept. CSE
SRM Valliammai Engineering College
Kattankulathur, India
anithar.cse@srmvalliammai.ac.in

Karthick S
Dept. CSE
SRM Valliammai Engineering College
Kattankulathur, India
17.karthick.03@gmail.com

Naveen Kumar S
Dept. CSE
SRM Valliammai Engineering College
Kattankulathur, India
naveenkumars081005@gmail.com

Nikhil B
Dept. CSE
SRM Valliammai Engineering College
Kattankulathur, India
nikhilbalaram03@gmail.com

Abstract: Printing services in educational institutions and public facilities are often managed through manual print shops, which can lead to long queues, limited service hours, and potential document privacy issues. This paper presents a QR-Based Secure Self-Service Printing Kiosk with Online Payment Integration to automate and secure the document printing process. The system allows users to upload PDF documents through a web interface, calculate page counts automatically, and complete digital payments using the Razorpay gateway. After successful payment, a unique one-time print code is sent to the user via email and used to authenticate the print job at the kiosk. The system is developed using the Flask web framework with SQLite for job management, while Ghostscript and CUPS enable automated document processing and printer control. The kiosk validates the print code to prevent unauthorized or duplicate printing. Experimental results show that the proposed system reduces waiting time, improves service efficiency, and provides a scalable solution for automated printing in educational campuses, libraries, and public service environments.

Keywords— Self-Service Printing, Secure Print Release, QR Authentication, Flask, Digital Payment, Printing Kiosk.

1. INTRODUCTION

The rapid growth of digital documents and online learning environments has significantly increased the demand for printing services in educational institutions, libraries, and public facilities. Despite this demand, many printing services are still operated through traditional print shops where users must physically visit the shop, submit their documents manually through pen drives or email, wait for operators to process the request, and make cash payments. This conventional approach often leads to long queues, delayed service, limited operating hours, and increased dependency on shop personnel. Additionally, manual handling of user documents may introduce privacy concerns and inefficiencies in managing print jobs.

With the advancement of web technologies, automation, and digital payment systems, self-service kiosks have emerged as an effective solution for improving service efficiency and user convenience. Self-service systems have already been widely adopted in applications such as Automated Teller Machines (ATMs), ticket vending machines, and smart vending systems. These technologies enable users to perform transactions independently without relying on human operators. Inspired by these developments, automated printing kiosks can provide a similar level of convenience by allowing users to upload documents, perform payments digitally, and retrieve printed copies directly from a kiosk-based printer.

Recent advancements in cloud computing, web applications, and payment gateway integration have enabled the development of secure and scalable remote printing systems. These systems allow users to upload documents online and execute printing tasks through centralized servers. However, many existing solutions rely heavily on continuous internet connectivity, cloud infrastructure, or manual print authorization mechanisms, which may introduce security vulnerabilities or operational limitations in certain environments.

To address these challenges, this paper proposes a **QR-Based Secure Self-Service Printing Kiosk with Online Payment Integration** that automates the entire printing workflow. The proposed system allows users to securely upload PDF documents through a web-based interface, automatically calculate the number of pages, and complete payments using a digital payment gateway. After successful payment confirmation, a unique one-time print code is generated and sent to the user via email. At the kiosk, users enter the code to authenticate and release the print job. The backend system validates the code, prevents duplicate printing, and automatically executes the print job using an integrated printing service.

The system is implemented using the Flask web framework, SQLite database for job management, Razorpay payment gateway for digital transactions, and Ghostscript for grayscale document processing. By combining web-based document submission, secure payment processing, and kiosk-based print

authentication, the proposed system provides a convenient, secure, and scalable solution for modern printing services. The remainder of this paper discusses related research, system design methodology, implementation details, and performance evaluation of the proposed printing kiosk system.

2. LITERATURE SURVEY

The demand for convenient and efficient printing services has increased significantly with the widespread use of digital documents in educational institutions, offices, and public environments. Traditional printing systems typically require users to visit print shops, transfer files using storage devices, and wait in queues for printing services. This manual process often leads to delays, inefficient job management, and limited service accessibility. To address these issues, researchers have explored remote printing and automated printing platforms that allow users to submit print requests digitally and collect printed documents later from designated locations [1].

One early solution to reduce queue time was the development of remote printing applications that allow users to upload documents through web or mobile interfaces. For example, the Print-OnTheGo system was developed using the MERN stack to enable users to send print requests to nearby print shops while specifying printer properties such as page count and print format. The system reduces waiting time and provides notifications when print jobs are ready for collection, thereby improving printing efficiency [1]. Similarly, mobile-based printing solutions using Android applications and cloud-based systems allow users to upload documents remotely and manage print queues digitally, reducing manual interaction between users and print operators.

With advancements in cloud computing, cloud-based printing frameworks have emerged to provide scalable and flexible printing services. These systems enable users to upload documents to centralized servers where print jobs are processed and delivered to available printers. Middleware solutions such as SmartPrint allow users to access multiple printers across a network without installing printer drivers and use intelligent printer allocation algorithms to optimize printer usage [2]. However, cloud-based solutions often rely heavily on stable internet connectivity and centralized infrastructure, which can limit their usability in certain environments.

Research has also focused on improving the design and usability of self-service terminals (SSTs) such as kiosks and vending machines. These systems are widely used in sectors including banking, ticketing, and automated retail due to their ability to provide services without human assistance. Studies on ticket vending machines emphasize user-centered design approaches that incorporate iterative testing and user feedback to ensure accessibility and usability for different user groups. Such approaches demonstrate how intuitive interfaces and automated workflows can improve the overall user experience in self-service environments [3].

Security and transaction management are also important aspects of self-service systems. Modern research proposes secure transaction frameworks that protect automated systems from cyber threats and unauthorized access. The AUTOTHING framework introduces a distributed transaction architecture designed for self-service terminals, utilizing multi-signature tokens and secure device-to-device communication channels to enhance system resilience against cyberattacks while maintaining operational efficiency [4].

Recent studies also explore advanced identity management and authentication mechanisms for automated service machines. For instance, a self-sovereign identity–based authentication framework has been proposed for automated teller machines (ATMs) to eliminate card-based authentication and improve transaction security. By leveraging blockchain-based decentralized identity systems, users can authenticate securely without relying on traditional card and PIN mechanisms, thereby improving privacy and reducing the risk of fraud [5].

Despite these advancements in remote printing systems, cloud printing services, and self-service kiosks, several limitations remain. Many existing systems depend on centralized cloud infrastructures, which may introduce latency and connectivity issues. Additionally, some platforms still require manual intervention from print shop operators, reducing the overall automation level. Security vulnerabilities and the lack of integrated payment and authentication mechanisms also remain challenges. Furthermore, many systems focus mainly on remote document submission or printer discovery rather than providing a complete automated printing workflow from document upload to secure print release [2], [4].

To address these limitations, the proposed system introduces a secure self-service printing kiosk with online payment integration. The system enables users to upload documents through a web interface, perform digital payments, and retrieve printed documents using a unique authentication code at a kiosk. By integrating web-based document management, secure transaction processing, and automated print release mechanisms, the system aims to reduce waiting time, improve operational efficiency, and provide a secure and convenient printing solution for environments such as universities, libraries, and offices.

3. METHODOLOGY

The proposed **QR-Based Secure Self-Service Printing Kiosk with Online Payment Integration** automates the entire printing workflow by combining document upload, digital payment processing, secure authentication, and automated printer control. The system is designed to minimize manual intervention while ensuring secure document handling and efficient print management.

The methodology consists of five major components:
(A) User Interface and Document Upload, (B) Payment Processing and cost calculation, (C) Backend System Architecture, (D) Secure Print Authentication, and (E) Automated Printing and Job Execution, (F) Workflow

The complete workflow is illustrated in **Fig. 1.**

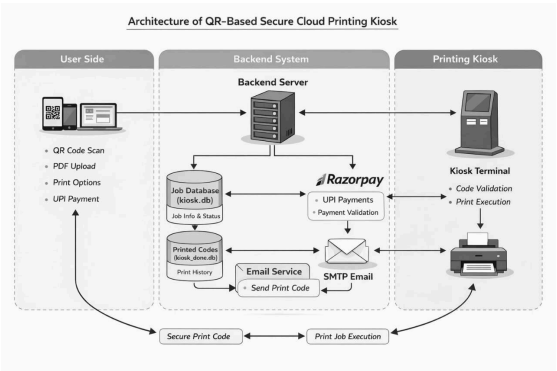

Fig 1 - System Architecture

A. User Interface and Document Upload

The system provides a **web-based interface developed using the Flask framework**, enabling users to upload documents and configure printing options through a web browser.The main functionalities of this module include:

- **PDF Document Upload:**
  Users can upload PDF documents through the web interface for printing.
- **Automatic Page Detection:**
  The system automatically detects the total number of pages in the uploaded document using a PDF processing library.
- **Print Configuration Options:**
  Users can select printing preferences such as: Black-and-white or color printing, Number of copies
- **Unique Job Identification:**
  Each uploaded document is assigned a **unique identifier generated using UUID**, ensuring that every print job is uniquely tracked within the system.
- **Secure File Storage:**
  Uploaded files are securely stored on the server with their unique job ID to prevent conflicts between multiple user requests.

B. Payment Processing and cost calculation

Once the document upload is completed, the system calculates the printing cost based on the number of pages and selected printing options.
The payment module includes the following components:

- **Cost Calculation Engine:**
  The system calculates the printing cost dynamically using predefined pricing rules for page count and print type.
- **Digital Payment Gateway Integration:**
  Payments are processed through the Razorpay payment gateway, which supports secure online transactions including UPI payments.
- **Payment Verification:**
  After the payment is completed, the backend server verifies the transaction status through the Razorpay API to confirm successful payment.
- **Generation of Secure Print Code:**
  Upon successful payment verification, the system generates a unique one-time print code that is linked to the specific print job.

C. Backend System Architecture

The backend system is responsible for managing document uploads, payment verification, job tracking, and communication with the kiosk terminal.
The backend architecture includes:

- **Flask Application Server:**
  The backend server is implemented using the Flask web framework, which manages all application logic and API communication.
- **Database Management:**
  The system uses SQLite databases to store and manage print job information.
- **Job Database (kiosk.db):**
  Stores important job-related information such as:
  - Job ID
  - User details
  - Uploaded file path
  - Page count
  - Payment status
  - Print code
  - Job status
- **Print History Database (kiosk_done.db):**
  Maintains records of printed jobs to prevent duplicate printing and maintain print history.
- **Email Notification Service:**
  An SMTP service is integrated to send the print code to the user after payment.

D. Secure Print Authentication

To ensure secure document handling, the system uses a code-based authentication mechanism at the kiosk terminal.
The authentication process includes:

- **Print Code Validation:**
  Users enter the received print code at the kiosk terminal.
- **Job Verification:**
  The kiosk communicates with the backend server to verify the code and retrieve the associated print job.
- **Security Checks:**
  The system performs multiple validation checks, including:
  1. Verification of payment status
  2. Confirmation that the job has not already been printed
  3. Validation of the entered print code
- **Prevention of Unauthorized Printing:**
  Only valid and paid jobs are allowed to proceed to the printing stage.

E. Automated printing and Job Execution

After successful authentication, the system automatically executes the printing process through the kiosk.
The automated printing process involves the following steps:

- **Job Retrieval:**
  The kiosk retrieves the document associated with the validated print code.
- **Document Processing:**
  If black-and-white printing is selected, Ghostscript is used to convert the document into grayscale format before printing.
- **Printer Communication:**
  The system uses the CUPS (Common UNIX Printing System) to send print commands to the connected printer.
- **Print Job Execution:**
  The document is printed automatically without requiring manual intervention.
- **Job Status Update:**
  Once printing is completed:
  1. The job status is updated to printed.
  2. The print code is stored in the print history database to prevent reuse.

F. Workflow Summary

The complete workflow of the system operates as follows:

1. The user scans the kiosk QR code or opens the web interface.
2. The user uploads a PDF document through the Flask web application.
3. The system detects the page count and calculates the printing cost.
4. The user completes payment through the Razorpay gateway.
5. A unique one-time print code is generated and sent to the user via email.
6. The user enters the print code at the kiosk terminal.
7. The kiosk validates the code and retrieves the associated document.
8. The printer executes the print job automatically through the CUPS printing system.
9. The system updates the job status and records the print history to prevent duplicate printing.

This automated workflow significantly reduces waiting time, enhances document security, and improves operational efficiency compared to traditional manual printing services.

4. RESULT AND DISCUSSION

The proposed **QR-Based Secure Self-Service Printing Kiosk with Online Payment Integration** was implemented and tested to evaluate the system's functionality, usability, and printing automation capabilities. The system was developed using the Flask web framework with SQLite database support and integrated with the Razorpay payment gateway for secure digital transactions. The experimental results demonstrate that the system successfully automates the entire printing workflow from document upload to print execution.

A. User Authentication and Account Management

The system begins with a **secure login interface**, where users can access the platform using their registered email address and password. The login page provides a simple and user-friendly interface that allows users to authenticate securely before accessing printing services. If the user does not have an account, they can create one through the registration option.

Once logged in, users can access their **profile management dashboard**, which allows them to view account information, update their password, and delete their account if required. The profile section also provides access to the user's **print history**, enabling them to track previously printed documents. This feature

enhances system usability and provides transparency for users regarding their printing activities.

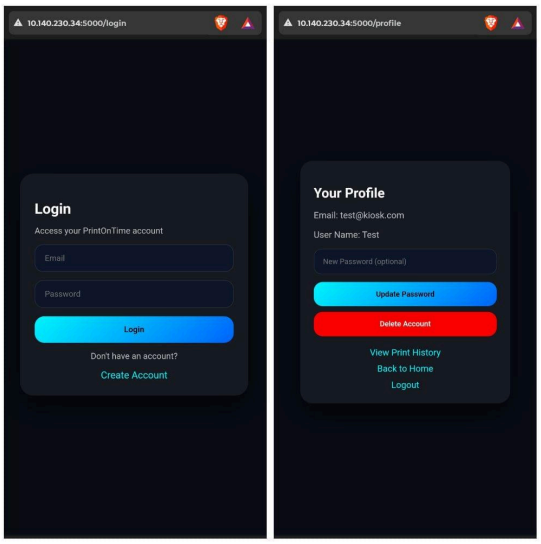

Fig.2 - Login and Profile Page

B. Document Upload and Print Configuration

After successful authentication, the user can upload a PDF document through the **document upload interface**. The platform provides a drag-and-drop style upload area that simplifies the process of submitting print jobs. Once the document is uploaded, the system automatically analyzes the PDF file and detects the number of pages.

The **print configuration module** allows users to select printing preferences such as black-and-white or color printing and adjust the number of copies required. The system dynamically calculates the printing cost based on these parameters and displays the total cost to the user before proceeding to payment. Additionally, the system provides an option to customize page selections for more flexible printing.

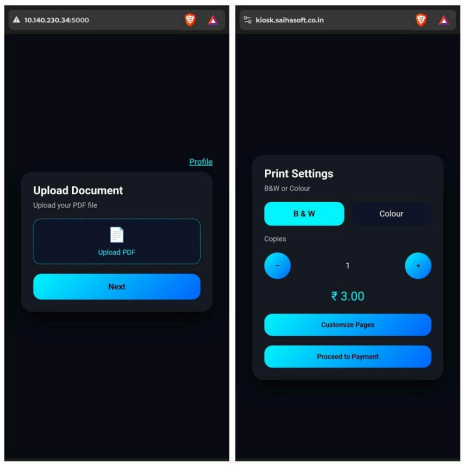

Fig.3 - Upload and Customize Page

C. Payment Processing

The system integrates the Razorpay payment gateway to enable secure online transactions. Users can complete the payment through various UPI-based payment applications such as Google Pay, PhonePe, or Paytm. The payment interface displays a QR code that users can scan using their preferred UPI application.

Once the payment is successfully completed, the system automatically verifies the transaction status and generates a unique six-digit print code. This code serves as a secure authentication token required to release the print job at the kiosk. The code is displayed on the confirmation page and also sent to the user's registered email address to ensure secure access.

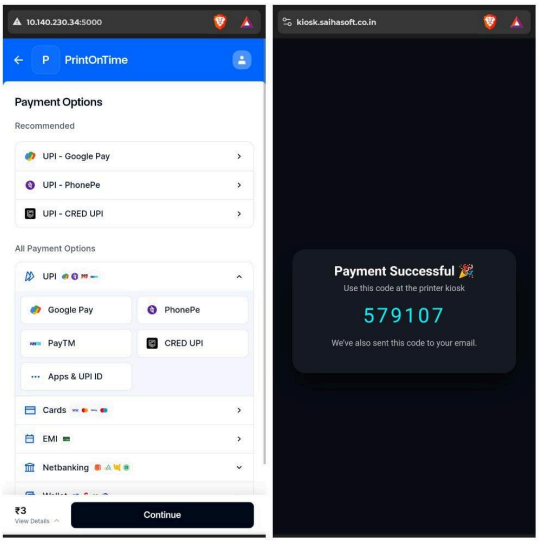

Fig. 4 - Payment Page

D. Secure Print Code Validation and Automated Printing

At the kiosk terminal, the user enters the generated print code through a keypad interface displayed on the screen. The system validates this code by checking the backend database to ensure that it corresponds to a valid and paid print job. If the code is verified successfully, the system retrieves the associated document from the server and displays a confirmation screen showing details such as the file name, printing mode, and number of copies, allowing the user to verify the print job before execution.

Once the print job is confirmed, the system automatically sends the document to the printer through the **CUPS printing system**. The connected printer processes the document and prints the output without requiring manual intervention. The experimental setup used a laptop as the kiosk terminal connected to an Epson printer, demonstrating that the system can be deployed using

standard computing hardware. The printed outputs confirm that the system accurately retrieves uploaded documents and executes printing successfully.

The experimental evaluation shows that the proposed system effectively automates the entire printing workflow while maintaining security and efficiency. The system reduces waiting time, enables secure authentication using one-time print codes, supports seamless digital payment integration through UPI, and ensures efficient job tracking while preventing duplicate printing. These results demonstrate that the proposed kiosk-based printing solution is suitable for deployment in environments such as universities, libraries, offices, and public service centers.

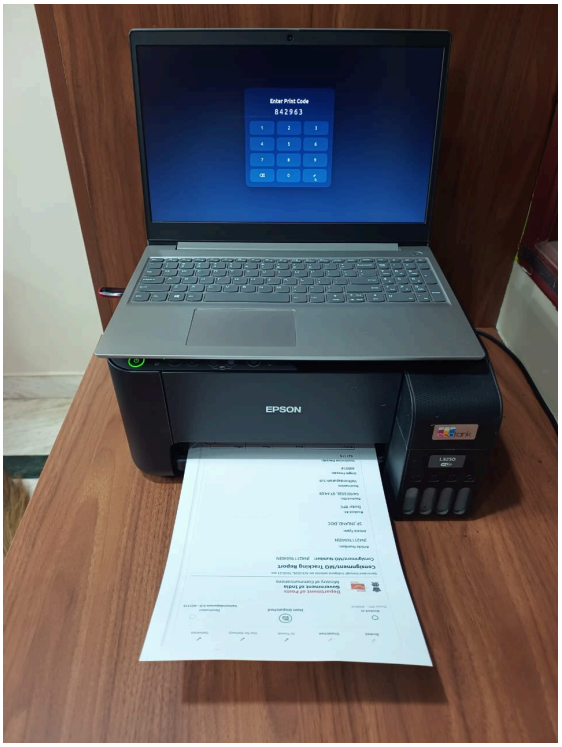

Fig. 5: Automated Printer

5. CONCLUSION

This paper presented a **QR-Based Secure Self-Service Printing Kiosk with Online Payment Integration** designed to automate the traditional document printing workflow and improve overall service efficiency. Conventional printing services in educational institutions and public facilities often rely on manual print shops, which can lead to long waiting times, limited service availability, and potential document privacy concerns. The proposed system addresses these challenges by providing a secure, automated, and user-friendly printing solution that integrates web-based document submission, digital payment processing, and kiosk-based print authentication.

The developed platform allows users to upload PDF documents through a web interface, automatically detect the number of pages, and configure printing preferences such as print mode and number of copies. The system calculates the printing cost dynamically and enables users to complete payments securely through the Razorpay payment gateway. After successful payment verification, a unique one-time print code is generated and delivered to the user, which serves as an authentication mechanism for releasing the print job at the kiosk terminal.

The backend system, implemented using the Flask web framework and SQLite database, manages document storage, job tracking, and payment verification. At the kiosk, the entered print code is validated against the database to ensure that only authorized and paid jobs are executed. Once verified, the system automatically sends the document to the connected printer through the CUPS printing service, ensuring a fully automated printing process without manual intervention.

Experimental evaluation demonstrates that the proposed system successfully automates the entire printing workflow while maintaining secure access and efficient job management. The solution reduces waiting time, improves operational efficiency, and ensures secure document handling through one-time print authentication. By integrating web-based document management, secure payment processing, and automated printing services, the system provides a scalable and cost-effective printing infrastructure suitable for deployment in universities, libraries, offices, and other public service environments.

6. REFERENCES

[1] M. A. Khan, S. Ahmed, and R. Ali, “Print-On-TheGo: A Remote Printing Application Using MERN Stack,” in Proc. IEEE Int. Conf. Smart Computing and Systems Engineering, 2024, pp. 1–6.

[2] A. Gupta, R. Sharma, and P. Singh, “AUTOTHING: A Secure Transaction Framework for Self-Service Things,” IEEE Access, vol. 10, pp. 78421–78432, 2022.

[3] L. Chen, Y. Wang, and H. Zhang, “A User-Centered Design Approach for Self-Service Ticket Vending Machines,” IEEE Transactions on Human-Machine Systems, vol. 52, no. 4, pp. 612–621, 2023.

[4] J. Kim and H. Lee, “Self-Sovereign Identity Empowered Automated Teller Machines,” IEEE Access, vol. 11, pp. 51234–51245, 2023.

[5] S. Patel and M. Shah, “Cloud-Based Remote Printing System for Smart Campus Environments,” in Proc. IEEE

Int. Conf. Cloud Computing Technology and Science, 2022, pp. 245–250.

[6] R. Kumar and P. Mehta, “SmartPrint: Middleware Architecture for Driverless Cloud Printing Services,” IEEE Internet of Things Journal, vol. 8, no. 14, pp. 11231–11240, 2021.

[7] T. Nguyen and D. Tran, “Mobile-Based Printing System Using Android and Cloud Services,” in Proc. IEEE Int. Conf. Mobile Computing and Applications, 2020, pp. 98–103.

[8] A. Singh and K. Verma, “Raspberry Pi-Based Wireless Printing System for Remote Document Access,” International Journal of Computer Applications, vol. 178, no. 6, pp. 15–20, 2019.

[9] H. Park and J. Choi, “Location-Aware Remote Printing Service for Campus Environments,” IEEE Transactions on Consumer Electronics, vol. 66, no. 2, pp. 210–218, 2020.

[10] V. Rao and S. Nair, “Queue-Based Online Printing Service System with Secure Token Authentication,” in Proc. IEEE Int. Conf. Smart Systems and Applications, 2021, pp. 120–125.